

1. 求积分 $I = \oint_C \frac{\bar{z}}{|z|} dz$ 值, 其中 C 为正向圆周.

(1) $|z| = 2$

$$z\bar{z} = |z|^2 \Rightarrow \frac{\bar{z}}{|z|} = \frac{|z|}{z}, I = \oint_C \frac{|z|}{z} dz$$

$$\therefore I = \oint_{|z|=2} \frac{2}{z} dz = 2 \cdot 2\pi i = 4\pi i$$

或 $z = 2e^{i\theta} (0 \leq \theta \leq 2\pi)$

$$I = \int_0^{2\pi} 2i d\theta = 4\pi i$$

(2) $I = \oint_{|z|=4} \frac{4}{z} dz = 4 \cdot 2\pi i = 8\pi i$

2. 沿指定曲线的正向计算下列求积的值.

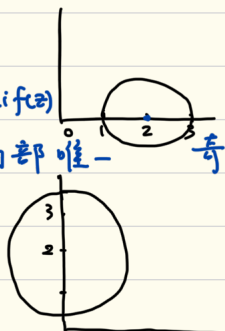
(1) $I = \oint_C \frac{e^z}{z-2} dz, C: |z-2|=1$

$$= \oint_C \frac{e^z}{z-2} dz$$

$$= 2\pi i e^2$$

$$\oint_C \frac{f(\zeta)}{\zeta-z} d\zeta = 2\pi i f(z)$$

$\zeta \in z$ 是 $\frac{f(\zeta)}{\zeta-z}$ 内部唯一奇点



(2) $I = \oint_C \frac{e^{iz}}{z^2+1} dz, C: |z-2i| = \frac{3}{2}$

奇点 $z = \pm i$. 其中 $z = i$ 包含在积分曲线内

$$I = \oint_C \frac{e^{iz}}{z^2+1} dz$$

$$= 2\pi i f(z)$$

$$= 2\pi i \cdot \frac{e^{i \cdot i}}{i+i}$$

$$= \pi e^{-1}$$

(3) $I = \oint_C \frac{1}{(z^2-1)(z^3-1)} dz, C: |z| = r < 1$

奇点: $1, -1, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

$$I = 0$$



(4) $I = \oint_C \frac{1}{(z^2+1)(z^2+\varphi)} dz, C: |z| = \frac{3}{2}$

奇点: $\pm i, \pm 2i$



被积函数在 C 内 $z = \pm i$ 处不解析. 在 C 内作正向圆周,

则函数在 C, C_1, C_2 围成的区域内解析.

$$\therefore \oint_C F(z) dz = \int_{C_1} F(z) dz + \int_{C_2} F(z) dz$$

$$\oint_{C_1} = \oint_{C_1} \frac{1}{\frac{(z^2+4)(z+i)}{z-i}} dz = 2\pi i \cdot \frac{1}{(i^2+\varphi)(i+i)} = \frac{\pi}{3}$$

$$\oint_{C_2} = \oint_{C_2} \frac{1}{\frac{(z^2+\varphi)(z-i)}{z+i}} dz = 2\pi i \cdot \frac{1}{[(-i)^2+\varphi](-i-i)} = -\frac{\pi}{3}$$

$$\therefore \oint_C = 0.$$

3(1) $I = \oint_{C=C_1+C_2} \frac{\cos z}{z^3} dz, C_1: |z|=2 \text{ 正向}; C_2: |z|=3 \text{ 负向}.$

$F(z)$ 在区域上解析.

$$\therefore I = 0.$$

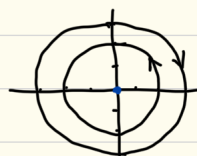
(2) $I = \oint_C \frac{e^z}{(z-a)^3} dz$, 其中 a 为 $|a| \neq 1$ 的任何复数. $C: |z|=1$ 为正向.

① 若 $|a| < 1$, a 为奇点

$$\text{原式} = \frac{2\pi i}{(3-1)!} (e^z)'' \Big|_{z=a}$$

$$= \pi e^a$$

$$\oint_C \frac{f(\zeta)}{(\zeta-z)^{n+1}} d\zeta = \frac{2\pi i}{n!} f^{(n)}(z)$$



② 若 $|a| > 1$, $F(z)$ 在区域上解析.

$$\therefore \oint_C = 0.$$

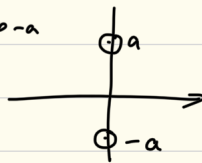
(3) $I = \oint_C \frac{z}{z^2 - a^2} dz$, 其中 $a \neq 0$ 为复数, C 为不经过 a 与 $-a$ 的简单闭曲线.

$$I = \oint_C \frac{z}{(z+a)(z-a)} dz$$

分别包含 a 和 $-a$

① 若 C 包含 a 和 $-a$, 在 C 内作正向圆周 C_1, C_2 , 则函数在

由 C, C_1^-, C_2^- 围成的区域内解析.



$$\begin{aligned} \therefore \oint_C &= \oint_{C_1} + \oint_{C_2} = \oint_{C_1} \frac{z}{z-a} dz + \oint_{C_2} \frac{z}{z+a} dz \\ &= 2\pi i \cdot \frac{a}{a+a} + 2\pi i \cdot \frac{-a}{-a-a} \\ &= 2\pi i \end{aligned}$$

② 若 C 包含 a , 不包含 $-a$.

$$I = \oint_C \frac{z}{z-a} dz = 2\pi i \cdot \frac{a}{a-a} = \pi i$$

③ 若 C 包含 $-a$, 不包含 a .

$$I = \oint_C \frac{z}{z+a} dz = \pi i$$

④ 若 C 不包含 a 和 $-a$.

$$I = 0.$$

4. 证明: $u = x^2 - y^2$ 和 $v = \frac{y}{x^2 + y^2}$ 都是调和函数, 但 $u + iv$ 不是解析函数

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial^2 u}{\partial x^2} = 2, \frac{\partial^2 u}{\partial y^2} = -2$$

$$\therefore \Delta u = 0$$

$$\begin{aligned} \frac{\partial v}{\partial x} &= y \cdot \left(-\frac{1}{(x^2+y^2)^2} \right) \cdot 2x = \frac{-2xy}{(x^2+y^2)^2}, \quad \frac{\partial^2 v}{\partial x^2} = \frac{(-2y) \cdot (x^2+y^2)^2 - (-2xy) \cdot 2(x^2+y^2) \cdot 2x}{(x^2+y^2)^4} = \frac{6x^2y - 2y^3}{(x^2+y^2)^3} \\ \frac{\partial v}{\partial y} &= \frac{1 \cdot (x^2+y^2) - y \cdot 2y}{(x^2+y^2)^2} = \frac{x^2 - y^2}{(x^2+y^2)^2}, \quad \frac{\partial^2 v}{\partial y^2} = \frac{-6xy + 2y^3}{(x^2+y^2)^3} \end{aligned}$$

$$\therefore \Delta v = 0$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -2y \neq -\frac{\partial v}{\partial x}$$

$\therefore u + iv$ 不是解析函数, 不满足 C-R 方程

5. 已知下列调和函数求解析函数 $f(z) = u + iv$

(1) $u = 2(x-1)y, f(z) = -i$

$$u_x = 2y, \quad u_y = 2(x-1) = -v_x$$

$$dv = v_x dx + v_y dy$$

$$= -u_y dx + u_x dy$$

$$\therefore v = \int u_x dy + \varphi(x)$$

$$= y^2 + \varphi(x)$$

$$v_x = \varphi'(x) = -2(x-1)$$

$$\varphi(x) = -x^2 + 2x + C$$

$$\therefore v = -x^2 + y^2 + 2x + C$$

$$\therefore f(z) = 2(x-1)y + (-x^2 + y^2 + 2x + C)i$$

$$f(z) = 2(x-1)y + [-(x-1)^2 + y^2 + C]i$$

$$\text{令 } x = z, y = 0.$$

$$f(z) = [-(z-1)^2 + C]i \Rightarrow C = 0.$$

$$\therefore f(z) = -(z-1)^2 i$$

$$(2) \quad v = \arctan \frac{y}{x}, \quad x > 0.$$

$$\frac{\partial u}{\partial x} = + \frac{\partial v}{\partial y} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = - \frac{\partial v}{\partial x} = \frac{y}{x^2 + y^2}$$

$$\therefore du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$u = \frac{1}{2} \ln(x^2 + y^2) + \varphi(y)$$

$$\frac{\partial u}{\partial y} = \frac{y}{x^2 + y^2} + \varphi'(y) = \frac{y}{x^2 + y^2}$$

$$\varphi'(y) = 0, \quad \varphi(y) = C.$$

$$\therefore u = \frac{1}{2} \ln(x^2 + y^2) + C$$

$$\therefore u + iv = \frac{1}{2} \ln(x^2 + y^2) + C + \arctan \frac{y}{x} i$$

$$\text{令 } x = z, y = 0,$$

$$f(z) = \ln z + C$$